Market Integration, Platformization, and Local Multipliers: The Spatial Reorganization of Value Capture

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July 17, 2026

Abstract

Using a standard local-income multiplier closure, we study how integration reallocates regional expenditure across channels with local retention. We let capacity raise platform retention without changing prices or demand. With no outside-funded retail spending, an endpoint-crossing condition yields ordered thresholds: both nominal and consumption-equivalent real income fall at low capacity, only the latter rises at intermediate capacity, and both rise at high capacity. When platform retention is below local-agent retention, any homothetic two-channel demand system with a strictly concave share-speed generates a unique capture-drag peak, located below one-half under symmetry. Integration and capacity have increasing differences in both log-income measures.

Keywords: market integration; digital platforms; local multipliers; value capture; spatial incidence

JEL codes: F15; L81; R11; R12

1 Introduction

Market integration can lower consumer prices while changing where transaction income is received. Spending in a region need not become income there: platform services, fulfillment, seller margins, and ownership returns may accrue elsewhere. Because retained income finances further local demand, this spatial incidence affects both first-round income and its propagation.

Evidence from rural China shows that e-commerce access can reduce living costs without producing comparable gains for local producers and workers (Couture et al., 2021). Online trade also changes regional consumption and real wages (Fan et al., 2018), while retail entry can generate price gains alongside local labor and profit effects (Atkin et al., 2018).

The local-multiplier literature studies how tradable-employment and fiscal shocks propagate through local activity (Moretti, 2010; Chodorow-Reich, 2019). We take that recursive income closure as a benchmark. Our narrower object is the incidence path created when integration endogenously reweights expenditure across channels with different local-income coefficients. This trades multi-region production and platform conduct for transparent finite-change and shape results.

In the benchmark without outside-funded retail spending, we establish two main results. First, under an explicit endpoint-crossing condition, a finite integration shock generates unique and ordered capacity thresholds separating joint losses, consumption-equivalent real income gains with falling nominal local income, and joint gains. Second, when platform transactions retain less local income than local-agent transactions, marginal capture drag has a unique interior maximum for any homothetic two-channel demand system whose induced share-speed is strictly concave; symmetry of that speed places the peak below a platform share of one-half, with CES as a closed-form specialization. A corollary shows that integration and organizational capacity exhibit increasing differences in log nominal local income and consumption-equivalent real income. These are conditional short-run incidence results. The consumption-equivalent real-income index is not full welfare, and the model does not determine platform pricing, capacity investment, or multi-region general equilibrium.

2 Model

Consider a small region L with a local-agent channel A and an outside platform channel P. Integration $z \in \mathbb{R}$ lowers $p_P(z) = \bar{p}_P e^{-z}$ and may raise autonomous local income $B(z) = \bar{B} e^{\varepsilon z}$.

Assumption 1 (Environment and demand)**.** Let $\alpha, \beta > 0$, $\alpha + \beta < 1$, $p_A, \bar{p}_P, \bar{B} > 0$, and $G, \varepsilon \geq 0$. Retail consumption is generated by a strictly increasing, linearly homogeneous two-channel aggregator with differentiable unit-expenditure function $P_M(p_A, p_P)$. Along the integration path, Hicksian demands are unique and interior, and the platform expenditure share $s(z) \in (0,1)$ satisfies $s'(z) > 0$.

A representative resident solves

$$\max_{M,N,Z \geq 0} M^\alpha N^\beta Z^{1-\alpha-\beta} \quad \text{s.t.} \quad P_M M + N + Z = Y. \tag{1}$$

Lemma 1 (Demand identities)**.** *Expenditures on* (M, N, Z) *are* $(\alpha Y, \beta Y, (1-\alpha-\beta)Y)$*. Moreover,*

$$s(z) = \frac{\partial \ln P_M}{\partial \ln p_P} \in (0,1), \qquad \frac{d \ln P_M}{dz} = -s(z), \qquad R = K \frac{Y}{P_M^\alpha}, \tag{2}$$

where $K = \alpha^\alpha \beta^\beta (1-\alpha-\beta)^{1-\alpha-\beta}$.

The index R is consumption-equivalent real income within this static consumption problem; it does not subtract the opportunity cost of labor supplied at the normalized wage.

The Supplement gives a CES specialization, but the main results require only the demand properties stated above. Platform transactions retain the local income share

$$\rho_P(\kappa) = \underline{\rho}_P + \kappa \delta_\rho, \qquad \delta_\rho = \overline{\rho}_P - \underline{\rho}_P > 0, \tag{3}$$

where $\kappa \in [0,1]$, $0 \leq \underline{\rho}_P < \overline{\rho}_P \leq 1$, and $\rho_A \in [0,1]$. Capacity summarizes local seller participation, fulfillment and related services, and organizational coordination that raise the local factor-and-ownership income content of platform transactions. It changes retention, not prices, preferences, or the share path.

Average local value capture and its derivatives are

$$\omega(z, \kappa) = (1-s)\rho_A + s\rho_P(\kappa), \tag{4}$$

$$\omega_z = -\Delta(\kappa) s'(z), \qquad \omega_\kappa = s\delta_\rho, \qquad \Delta(\kappa) = \rho_A - \rho_P(\kappa). \tag{5}$$

The benchmark condition $\Delta(\kappa) > 0$ is imposed only where stated.

The outside-good price and local wage are one. A competitive nontradable sector has technology $N = L_N$, so its revenue βY is local labor income. Each channel dollar generates ρ_c dollars of local factor and locally owned business income. Outside-funded retail spending G follows the same channel shares. Local income therefore satisfies

$$Y = B + \beta Y + \omega(\alpha Y + G). \tag{6}$$

Lemma 2 (Local-income equilibrium)**.** *The income map is a contraction and has the unique positive fixed point*

$$Y(z,\kappa,G)=\frac{B(z)+\omega(z,\kappa)G}{D(z,\kappa)},\quad D=1-\beta-\alpha\omega=d_A+\alpha s\Delta(\kappa)>0,\quad d_A=1-\beta-\alpha\rho_A>0. \tag{7}$$

The standard benchmark multipliers are

$$m_B=\frac{\partial Y}{\partial B}=\frac{1}{D},\qquad \mathcal{M}_G=\frac{\partial Y}{\partial G}=\frac{\omega}{D}. \tag{8}$$

They interpret propagation; the results below concern how integration changes the retention-weighted denominator.

3 Results

Set $G=0$, take $z_1>z_0$, and write $s_j=s(z_j)$, $B_j=B(z_j)$, $P_{Mj}=P_M(z_j)$, and

$$D_j(\kappa)=d_A+\alpha s_j\Delta(\kappa),\quad \mathcal{B}=\frac{B_1}{B_0},\quad \mathcal{P}=\frac{P_{M1}}{P_{M0}},\quad \mathcal{D}(\kappa)=\frac{D_1(\kappa)}{D_0(\kappa)}. \tag{9}$$

Assumption 1 implies $s_1>s_0$ and $0<\mathcal{P}<1$.

Proposition 1 (Finite-change capacity thresholds)**.** *The finite income ratios and the capacity response of the capture wedge are*

$$\frac{Y_1}{Y_0}=\frac{\mathcal{B}}{\mathcal{D}(\kappa)},\qquad \frac{R_1}{R_0}=\frac{\mathcal{B}}{\mathcal{D}(\kappa)\mathcal{P}^\alpha},\qquad \frac{\partial\ln\mathcal{D}}{\partial\kappa}=-\frac{\alpha\delta_\rho d_A(s_1-s_0)}{D_1D_0}<0. \tag{10}$$

Suppose

$$\mathcal{D}(0)>q_R^F\equiv\frac{\mathcal{B}}{\mathcal{P}^\alpha}>q_Y^F\equiv\mathcal{B}>\mathcal{D}(1). \tag{11}$$

Then unique thresholds $0<\kappa_R^F<\kappa_Y^F<1$ are given by $\kappa_R^F=\kappa(q_R^F)$, $\kappa_Y^F=\kappa(q_Y^F)$, where

$$\kappa(q)=\frac{1}{\delta_\rho}\left[\Delta_0-\frac{(q-1)d_A}{\alpha(s_1-qs_0)}\right],\qquad \Delta_0=\rho_A-\underline{\rho}_P. \tag{12}$$

Below κ_R^F, nominal local income and consumption-equivalent real income fall. Between the thresholds, nominal local income falls and consumption-equivalent real income rises. Above κ_Y^F, both rise. At κ_R^F, consumption-equivalent real income is unchanged and nominal local income falls. At κ_Y^F, nominal local income is unchanged and consumption-equivalent real income rises.

Capacity leaves access prices and expenditure shares unchanged; it weakens only the finite denominator response. Consumption-equivalent gains can therefore appear before nominal local-income gains.

Corollary 1 (Integration–capacity increasing differences)**.** *When $G=0$, at every interior (z,κ),*

$$\frac{\partial^2\ln Y}{\partial z\,\partial\kappa}=\frac{\partial^2\ln R}{\partial z\,\partial\kappa}=\frac{\alpha\delta_\rho d_A s'(z)}{D^2}>0. \tag{13}$$

Thus both log-income measures have strict increasing differences in integration and organizational capacity. The equality for R follows because capacity does not enter P_M.

For marginal integration changes, write the demand-induced share speed as $s'(z) = g(s)$ and define

$$\Lambda_g(s,\kappa) = \frac{\alpha\Delta(\kappa)g(s)}{d_A + \alpha\Delta(\kappa)s}. \tag{14}$$

Then

$$\frac{d\ln Y}{dz} = \varepsilon - \Lambda_g, \qquad \frac{d\ln R}{dz} = \varepsilon + \alpha s - \Lambda_g. \tag{15}$$

Proposition 2 (Capture drag under a general share-speed)**.** *Fix κ with $\Delta(\kappa) > 0$. Suppose g, independent of κ, satisfies*

$$g \in C^2([0,1]), \quad g(0) = g(1) = 0, \quad g(s) > 0 \textit{ for } s \in (0,1), \quad g''(s) < 0 \textit{ for } s \in (0,1). \tag{16}$$

Then $\Lambda_g(\cdot,\kappa)$, extended to $[0,1]$, is strictly single-peaked with a unique interior maximizer $s^*(\kappa)$. If $g(s) = g(1-s)$, then $s^*(\kappa) < 1/2$. At interior capacity levels for which $\Delta(\kappa) > 0$,

$$\frac{\partial s^*}{\partial\kappa} > 0, \qquad \frac{\partial\Lambda_g^{\max}}{\partial\kappa} < 0. \tag{17}$$

The endpoint conditions say that spending cannot reallocate rapidly when the platform is almost absent or almost universal; strict concavity makes that reallocation speed rise and then fall without imposing CES. Under symmetry, $g'(1/2) = 0$, while the retention-weighted denominator already rises with platform penetration, so capture drag must peak before one-half. Higher organizational capacity narrows the retention gap, directly lowering the peak and shifting it right because greater platform penetration is then required to generate the strongest leakage pressure.

Figure 1 illustrates the non-CES share-speed result and the finite capacity crossings.

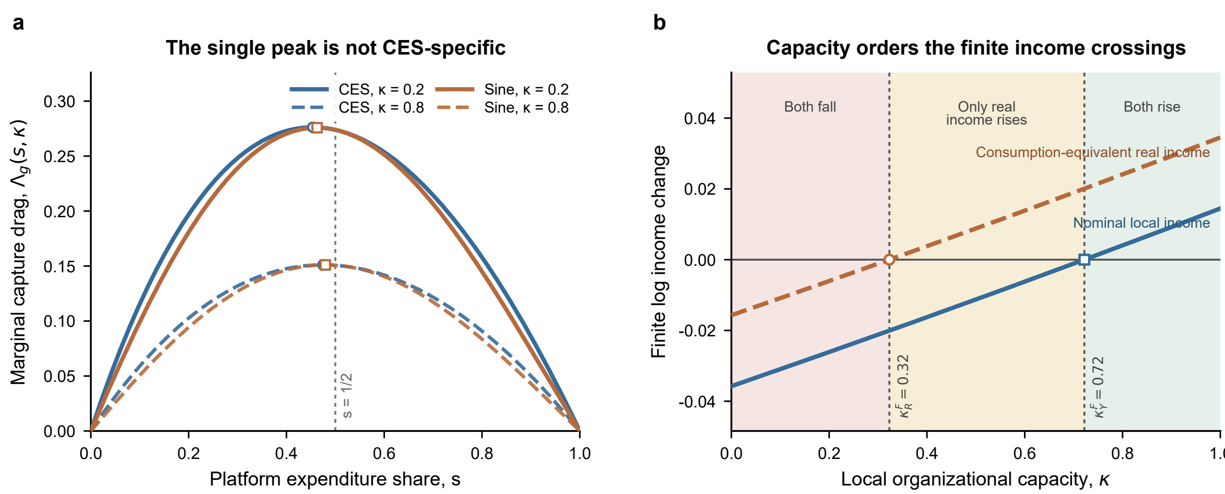

Figure 1: Capture drag and finite capacity thresholds. **(a)** CES $g(s) = 3s(1-s)$ and non-CES $g(s) = 0.75\sin(\pi s)$ are shown at $\kappa = 0.2$ and 0.8; both have a unique pre-half peak, and capacity moves it right while lowering it. **(b)** Finite log-income changes cross zero at $\kappa_R^F = 0.32 < \kappa_Y^F = 0.72$. Panel **(b)** uses the CES specialization with $\alpha = 0.35$, $\beta = 0.25$, $a = 0.4$, $\eta = 4$, $\varepsilon = 0.08$, $\rho_A = 0.8$, $(\underline{\rho}_P, \overline{\rho}_P) = (0.1, 0.6)$, and $(z_0, z_1) = (-0.75, -0.30)$. Illustration, not calibration.

4 Conclusion

Market integration affects local income through access and the spatial retention of transaction value. In the benchmark without outside-funded retail spending, the endpoint-crossing condition yields ordered capacity thresholds separating joint losses, consumption-equivalent real income gains with nominal local-income losses, and joint gains. When platform transactions retain less local income than local-agent transactions, the intermediate capture-drag peak does not require CES: it follows from a strictly concave demand-induced share-speed and precedes one-half under symmetry. In the same $G = 0$ benchmark, a corollary establishes increasing differences in both log-income measures. These results apply to a short-run small-region closure with exogenous prices, capacity, and retention schedules. The consumption-equivalent real-income index is not full welfare; platform pricing, capacity investment, production relocation, and multi-region feedback remain outside the model.

Data availability

No data were used for the research described in this article.

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